

System Design Document (SDD)

Physiological Digital Twin Berbasis CAPAR

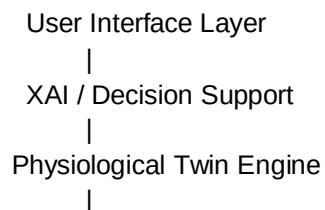
Dokumen ini menjelaskan desain teknis sistem Physiological Digital Twin berbasis Context-Aware Personalized Autonomic Recovery (CAPAR). SDD menjadi acuan implementasi bagi software engineer, data scientist, dan konseptor fisiologi.

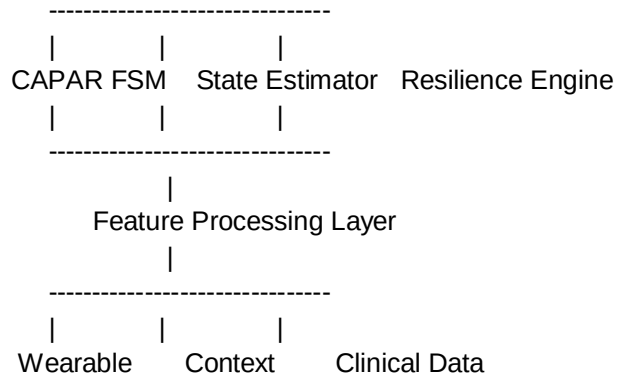
1. System Architecture Overview

Arsitektur sistem terdiri dari:

1. Data Acquisition Layer
 - Wearable sensor
 - Clinical database
 - Context input
2. Data Governance Layer
 - Data quality
 - Provenance
 - Privacy
 - Validation
3. Physiological Intelligence Layer
 - Signal processing
 - Feature extraction
 - Baseline engine
 - CAPAR engine
 - State estimation
4. Application Layer
 - Dashboard
 - XAI
 - Decision support
 - Reporting

2. High Level Architecture





3. Component Design

A. Data Acquisition Component

Input:

- HR
- RR/IBI
- HRV
- DFA alpha
- ACC
- Sleep
- EMA
- Clinical parameters

B. Preprocessing Component

Function:

- artifact removal
- filtering
- missing data handling
- quality scoring

C. Feature Engine

Menghasilkan:

- HR mean
- HR slope
- RMSSD
- SDNN
- pNN50
- LF/HF
- DFA alpha1

- local variance
- recovery slope
- TTR
- relapse index

D. CAPAR Engine

Fungsi:

- context normalization
- deviation calculation
- persistence detection
- recovery analysis
- relapse detection
- state transition

E. Digital Twin Engine

Mengestimasi:

$X(k)=$

[Vulnerability,
Cardiac Reserve,
Autonomic Reserve,
Recovery Capacity,
Regulation Stability]

4. Data Flow Design

Step 1:

Sensor menghasilkan raw physiological signal.

Step 2:

Preprocessing membersihkan dan memberi quality score.

Step 3:

Feature extraction menghasilkan physiological features.

Step 4:

Context engine menentukan expected response.

Step 5:

CAPAR engine menghitung deviation dan trajectory.

Step 6:

State estimator memperbarui digital twin.

Step 7:

Output:

- risk
- recovery trajectory
- phenotype
- XAI

5. Mathematical Model Design

State space representation:

$$z(k+1)=f(z(k),u(k),d(k))+w(k)$$

$$y(k)=h(z(k),u(k))+v(k)$$

$$X(k)=g(z(k),c(k),B_{\text{personal}},e(k))$$

$$s(k+1)=T(s(k),D(k),P(k),R(k))$$

Variable:

$u(k)$:

activity, context, stress, sleep, medication

$d(k)$:

noise, unobserved load, environment

$c(k)$:

risk factor, examination, laboratory

$z(k)$:

latent physiological state

$X(k)$:

resilience state

$s(k)$:

CAPAR event state

6. CAPAR Finite State Machine Design

State:

Baseline Mature

|
v

No Deviation

|
v

Deviation

|
v

Persistent Deviation

|
v

Recovery Start

|
v

Full Recovery

|
v

Baseline Update

Relapse:

Recovery Start -> Persistent Deviation

7. Database Design

Tabel utama:

Participant
- participant_id

- demographic
- clinical profile

Sensor_Data

- timestamp
- HR
- RR
- HRV
- ACC

Context_Data

- activity
- sleep
- stress
- medication

Feature_Table

- extracted features
- quality score

Episode_Table

- onset
- peak
- recovery
- relapse

Twin_State_Table

- resilience score
- latent state
- confidence

Explanation_Table

- evidence
- uncertainty

8. API Design

API utama:

POST /sensor/upload

Input wearable data

POST /clinical/update
Input clinical information

GET /twin/state
Mengambil current digital twin state

GET /episode/history
Mengambil episode trajectory

GET /xai/explanation
Mengambil explanation output

POST /model/update
Update model parameter

9. Algorithm Design

Algorithm pipeline:

1. Receive data
2. Validate quality
3. Filter signal
4. Extract feature
5. Determine context
6. Calculate personal z-score
7. Calculate deviation score
8. Update CAPAR FSM
9. Estimate resilience state
10. Generate explanation

10. Security and Governance Design

Requirement:

- encrypted storage
- authentication
- audit trail
- model versioning
- data provenance
- uncertainty reporting
- human oversight

11. Testing Design

Unit Testing:

- feature calculation
- state transition

Integration Testing:

- sensor to engine

Validation Testing:

- ECG comparison
- EMA ground truth
- expert annotation

Performance Testing:

- latency
- scalability
- reliability

12. Deployment Architecture

Edge Layer:

- wearable communication
- preprocessing ringan

Cloud Layer:

- database
- model inference
- analytics

Application Layer:

- web dashboard
- mobile application

13. Developer Implementation Guideline

Developer harus memastikan:

1. Setiap feature mempunyai traceability.
2. Setiap state mempunyai definisi fisiologi.
3. Threshold dapat dikonfigurasi.

4. Model dapat diperbarui.
5. Output memiliki confidence.
6. XAI tersedia untuk setiap keputusan.

14. Kesimpulan

System Design Document ini menerjemahkan konsep Physiological Digital Twin CAPAR menjadi rancangan perangkat lunak yang dapat diimplementasikan. Sistem tidak hanya melakukan anomaly detection, tetapi melakukan state estimation, resilience modeling, trajectory analysis, dan explainable physiological decision support.